Introduction to Econometrics
Week 3: Hypothesis testing part 1
(Hayashi section 1.4)

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Based on Rob Alessie lecture notes

Organization of the lecture

Hypothesis Testing **under Normality**

- Normally Distributed Error Terms
- Testing Hypotheses about Individual Regression Coefficients
- Decision Rule for the t-Test
- Confidence Interval
- p-Value

What is inference?

- Inference is one of the main idea of the exercise we are doing in this class
- We want to be able to test hypothesis about the parameters of the model describing the underlying population
- So, we want to be able to make statements about the *true* values of the $\beta's$
- In order to test hypothesis we need to construct a *test statistic* - expression that we can compute in our sample
- Then we need to know how this statistic behaves - so find its sample distribution
- Recall that the *estimates* differ across random samples from the underlying population - they are realizations of a random variable which is defined by the formulas for *estimators*

Hypothesis Testing under Normality

- Recall the regression model

$$y_i = \beta_1 + \beta_2 x_{i2} + \ldots + \beta_K x_{iK} + \varepsilon_i, \quad i = 1, \ldots, n \tag{1}$$

- Very often, the economic theory that motivated the regression equation also specifies the values that the regression coefficients should take.
- Example: Cobb-Douglas production function with constant returns to scale:

$$ln(Y_i) = \gamma_1 + \beta_1 ln(L_i) + \beta_2 ln(K_i) + \varepsilon_i, \quad i = 1, \ldots, n \tag{2}$$

where Y_i denotes output, L_i labor input, and K_i capital. If production technology exhibits **constant returns to scale**, then $\beta_1 + \beta_2 = 1$.

Hypothesis Testing under Normality

- The unbiasedness property of OLS ($E(\mathbf{b}|\mathbf{X}) = \boldsymbol{\beta}$) guarantees that (cf equation (2))

$$E(b_1 + b_2) = \beta_1 + \beta_2 = 1$$

  if the restriction of constant returns to scale is true.
- However $b_1 + b_2$ is not exactly equal to 1 for a particular sample at hand (due to the sampling error).
- Obviously, we cannot conclude that the restriction is false just because the estimate $b_1 + b_2$ differs from 1.
- To decide whether the sampling error

$$b_1 + b_2 - 1$$

  is too large for the restriction to be true, we need to construct from the sampling error some **test statistic** whose probability distribution is known given the truth of the hypothesis.
- In the language of hypothesis testing, the restriction to be tested (such as $\beta_1 + \beta_2 = 1$) is called the **null hypothesis**:

$$H_0 : \beta_1 + \beta_2 = 1$$

Normally distributed error terms

Assumption 1.5 (normality of the error term): *The distribution of* $\varepsilon|\mathbf{X}$ *is jointly normal.*

Normal distribution has several properties:

- 1. The distribution (density function) depends only on the mean and the variance
  - Once the mean and the variance are known, we can write down the density function
  - If the distribution conditional on $\mathbf{X}$ is normal, the mean and the variance can depend on $\mathbf{X}$
  - If neither the mean nor the variance depend on $\mathbf{X}$ the marginal (i.e. unconditional) distribution is the same normal distribution

Normally distributed error terms

- 2. In general, if two random variables are independent, than they are uncorrelated, but the converse is not true
  Example: Consider two random variables X and Y where $Y = X^2$. Moreover, assume that $E(X) = E(X^3) = 0$.
  Then since: $E(XY) = E(X^3) = E(X)E(Y) = 0 \Rightarrow cov(X, Y) = 0$ the variables X and Y are uncorrelated
  But if $Y = X^2$, then X and Y are clearly not independent: once we know X, we know Y.
  - However, it can be shown that if two random variables X and Y are jointly normally distributed, the converse is also true, so that independence and a lack of correlation are equivalent.
  - This carries over to conditional distributions: if two random variables are jointly normal and uncorrelated conditional on X, then they are independent conditional on X.

Normally distributed error terms

- 3. A linear function of random variables that are jointly normal is itself normally distributed.
  - This also carries over to conditional distributions
  - If the distribution of $\mathbf{u} = (u_1, \ldots, u_n)'$ conditional on $\mathbf{X}$ is normal, then $\mathbf{Au}$, where the elements of matrix $\mathbf{A}$ are functions of $\mathbf{X}$, is normal conditional on $\mathbf{X}$:

$$\left.\begin{array}{c} \mathbf{u}|\mathbf{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \\ \mathbf{A} \text{ is a function of } \mathbf{X} \end{array}\right\} \Rightarrow \mathbf{Au}|\mathbf{X} \sim N(\mathbf{A}\boldsymbol{\mu}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}')$$

Normally distributed error terms

Assumptions 1.5 thanks to features 1-3 grants us the following two features which we will exploit to deliver the test statistic:

- 1. The mean and the variance of the distribution of $\varepsilon|\mathbf{X}$ are already specified in assumptions 1.2 and 1.4

$$\varepsilon|\mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n) \tag{3}$$

  Thus the distribution of $\varepsilon|\mathbf{X}$ does not depend on $\mathbf{X}$

  It follows that ε and $\mathbf{X}$ are *independent* and therefore $\varepsilon \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$

Normally distributed error terms

- 2. From the definition of sampling error of $\mathbf{b}$, we know it is linear in ε given $\mathbf{X}$:

$$\begin{aligned} \mathbf{b} - \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} - \beta = \\ = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\beta + \varepsilon) - \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon \end{aligned} \tag{4}$$

  Since ε is normal given $\mathbf{X}$ (per assumption 1.5) so is the sampling error (given the properties of normal distribution discussed above):

$$\mathbf{b} - \beta | \mathbf{X} \sim N(\mathbf{0}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1}) \tag{5}$$

Testing Hypotheses about Individual Regression Coefficients

- Consider the following *Null Hypothesis* on the k-th regression coefficient
$$H_0 : \beta_k = \bar{\beta}_k$$
Here, $\bar{\beta}_k$ is some known value specified by the null hypothesis.
- We wish to test H_0 against the *Alternative hypothesis* (two sided test)
$$H_1 : \beta_k \neq \bar{\beta}_k$$
at a significance level α.
- Significance level α=maximum accepted probability of a type I error: $P(\text{reject } H_0 | H_0 \text{ true})$

  In other words significance level is the amount of evidence that needs to be collected to accept that the event is unlikely to have happened by chance

Testing Hypotheses about Individual Regression Coefficients

- Looking at the k-th component of the sampling error (5) and imposing the restriction of the null ($\beta_k = \bar{\beta}_k$), we obtain

$$b_k - \bar{\beta}_k | \mathbf{X} \sim N(0, \sigma^2 (\mathbf{X}'\mathbf{X})^{-1}_{kk}) \tag{6}$$

  where $(\mathbf{X}'\mathbf{X})^{-1}_{kk}$ is the (k,k) element of $(\mathbf{X}'\mathbf{X})^{-1}$.
- Standardizing $b_k - \bar{\beta}_k$ gives

$$z_k \equiv \frac{b_k - \bar{\beta}_k}{\sqrt{\sigma^2 (\mathbf{X}'\mathbf{X})^{-1}_{kk}}} \sim N(0,1) \tag{7}$$

Testing Hypotheses about Individual Regression Coefficients

- Suppose for the moment that σ^2 is known (obviously this is an unrealistic assumption)
- Then the statistic z_k has some desirable properties:
  1. its value can be calculated from the sample
  2. the conditional distribution $z_k|\mathbf{X} \sim N(0,1)$ does not depend on $\mathbf{X} \Rightarrow$ z_k and $\mathbf{X}$ are independently distributed and the unconditional distribution $z_k \sim N(0,1)$ is the same as the conditional one
  3. The distribution of z_k is known and does not depend on unknown *nuisance parameters*
  4. Using z_k, we can determine whether or not the sampling error $b_k - \bar{\beta}_k$ is too large.

Testing Hypotheses about Individual Regression Coefficients

- However σ^2 is typically not known.
- So replace in equation (7) the nuisance parameter σ^2 by its OLS estimate $s^2 = \mathbf{e}'\mathbf{e}/(n-K)$.
- In this way we obtain the following test statistic (t-ratio):

$$t_k \equiv \frac{b_k - \bar{\beta}_k}{\sqrt{\widehat{Var}(\mathbf{b}|\mathbf{X})_{kk}}} \equiv \frac{b_k - \bar{\beta}_k}{\sqrt{s^2(\mathbf{X}'\mathbf{X})^{-1}_{kk}}} \tag{8}$$

Proposition 1.3 (distribution of the t-ratio): *Suppose Assumptions 1.1,..,1.5 hold. Under the null hypothesis $H_0 : \beta_k = \bar{\beta}_k$, the t-ratio defined as in equation (8) is distributed as $t_{(n-K)}$ (the t-distribution with n-K degrees of freedom).*

Testing Hypotheses about Individual Regression Coefficients

- To prove this proposition, first notice a few facts:
  - *Fact 1:* If
    1. $x \sim N(0,1)$
    2. $y \sim \chi^2(m)$
    3. x and y are independent

    then the ratio $\frac{x}{\sqrt{y/m}}$ has the t-distribution with m degrees of freedom
  - *Fact 2*: If $\mathbf{x} \sim N(\mathbf{0}, \mathbf{I}_n)$ and $\mathbf{A}$ is symmetric and idempotent, then $\mathbf{x}'\mathbf{A}\mathbf{x} \sim \chi^2(trace(\mathbf{A}))$.
  - *Fact 3*: If $\mathbf{x}$ and $\mathbf{y}$ are independently distributed, then so are $\mathbf{f}(\mathbf{x})$ and $\mathbf{g}(\mathbf{y})$.
- We will also need the following proposition:
  Proposition 1.1.d: Under assumptions 1.1-1.4 $Cov(\mathbf{b}, \mathbf{e}|\mathbf{X}) = \mathbf{0}$ where $\mathbf{e} = \mathbf{y} - \mathbf{X}\mathbf{b}$

Testing Hypotheses about Individual Regression Coefficients

Proof proposition 1.3

- We can rewrite equation (8) as follows:

$$
\begin{aligned}
t_k &= \frac{b_k - \bar{\beta}_k}{\sqrt{\sigma^2 (\mathbf{X}'\mathbf{X})^{-1}_{kk}}} \sqrt{\frac{\sigma^2}{s^2}} = \frac{z_k}{\sqrt{\frac{s^2}{\sigma^2}}} \\
&= \frac{z_k}{\sqrt{\frac{\mathbf{e}'\mathbf{e}/(n-K)}{\sigma^2}}} = \frac{z_k}{\sqrt{\frac{q}{n-K}}}
\end{aligned} \tag{9}
$$

where $q = \frac{\mathbf{e}'\mathbf{e}}{\sigma^2}$

We know that $z_k \sim N(0,1)$. We will show:

(1) $q|\mathbf{X} \sim \chi^2(n-K)$

(2) Two random variables z_k and q are independent conditional on $\mathbf{X}$

Then, due to fact 1, t_k (cf. equation (9)) is distributed as t with n-K degrees of freedom. Then we have proven proposition 1.1.3.

Testing Hypotheses about Individual Regression Coefficients

NTS that (1) and (2) hold:
(1) Since $\mathbf{e}'\mathbf{e} = \varepsilon'\mathbf{M}\varepsilon$ (see material for week 2), we have

$$q = \frac{\mathbf{e}'\mathbf{e}}{\sigma^2} = \frac{\varepsilon'}{\sigma}\mathbf{M}\frac{\varepsilon}{\sigma}$$

where $\mathbf{M} = \mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ is symmetric and idempotent. We already know (proposition 1.2) that

$$trace(\mathbf{M}) = n - K$$

Also, due to equation (3)

$$\varepsilon/\sigma|\mathbf{X} \sim N(0, \mathbf{I}_n)$$

Due to fact 2, $q|\mathbf{X} \sim \chi^2(trace(\mathbf{M}))$ or $q|\mathbf{X} \sim \chi^2(n-K)$ (so (1) holds)

Testing Hypotheses about Individual Regression Coefficients

(2) Both $\mathbf{b} = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon$ and $\mathbf{e} = \mathbf{M}\varepsilon$ are linear functions of ε. So:

- $\mathbf{b}$ and $\mathbf{e}$ are jointly normal distributed conditional on $\mathbf{X}$
- From proposition 1.1.d, we know that $\mathbf{b}$ and $\mathbf{e}$ are uncorrelated conditional on $\mathbf{X}$

These two properties imply $\mathbf{b}$ and $\mathbf{e}$ are independently distributed conditional on $\mathbf{X}$

Notice that (a) z_k is a function of $\mathbf{b}$ and (b) q a function of $\mathbf{e}$

So z_k and q are independently distributed conditional on $\mathbf{X}$ (cf. fact 3). (so (2) holds)

q.e.d.

Decision rule of the t-test

Step 1: Formulate H_0 and H_1, e.g. $H_0 : \beta_k = \bar{\beta}_k$ and $H_1 : \beta_k \neq \bar{\beta}_k$ (two sided test).

Step 2: Specify the significance level (size) α, Typically, one chooses either $\alpha = 0.10, 0.05, 0.01$.

Step 3: Given the hypothesized value, $\bar{\beta}_k$, of β_k, form the t-ratio as in (8).

Step 4: Go to the t-table and look up the **critical value**, $t_{\alpha/2}(n-K)$.
Find the **critical value** such that the area in the t distribution to the right of $t_{\alpha/2}(n-K)$ is $\alpha/2$, see figure 1.3.
If $n-K$ is large, then the student t distribution is very similar to the standard normal distribution.
If $n-K = 900$ and $\alpha = 0.05$ then $t_{\alpha/2}(n-K) = 1.96$
Then, since the t distribution is symmetric around 0 (see figure 1.3)

$$P(-t_{\alpha/2}(n-K) < t_k < t_{\alpha/2}(n-K)) = 1-\alpha$$

Step 5: Do not reject H_0 if

$$-t_{\alpha/2}(n-K) < t_k < t_{\alpha/2}(n-K)$$

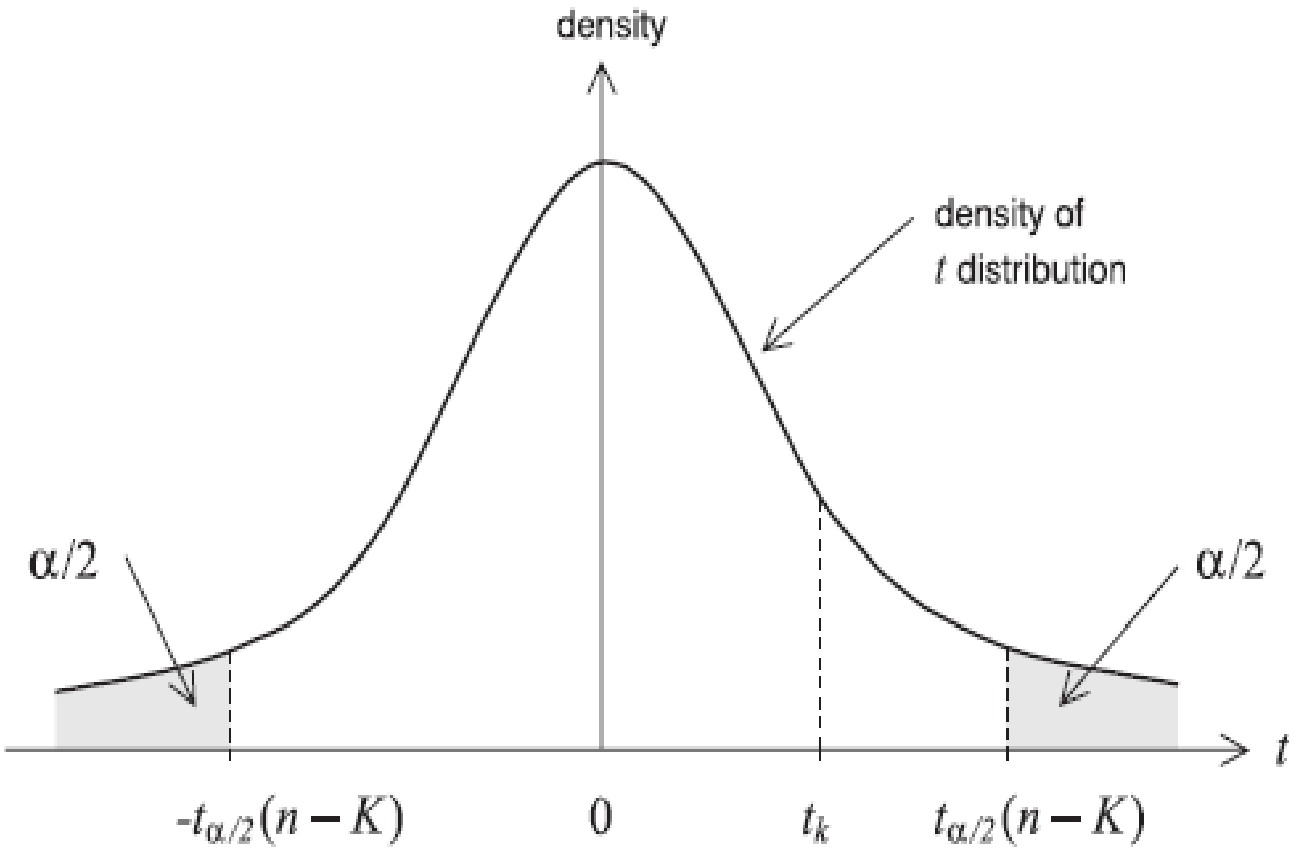

Figure 1.3: t Distribution

Confidence interval

- *Step 5* can also be stated in terms of b_k and $SE(b_k)$ (by definition of the test statistic).
- Since t_k is as in (8), you do not reject H_0 whenever

$$-t_{\alpha/2}(n-K) < \frac{b_k - \bar{\beta}_k}{SE(b_k)} < t_{\alpha/2}(n-K)$$

or

$$b_k - t_{\alpha/2}(n-K)SE(b_k) < \bar{\beta}_k < b_k + t_{\alpha/2}(n-K)SE(b_k)$$

- Therefore, we do not reject H_0 if and only if the hypothesized value $\bar{\beta}_k$ falls in the interval presented above
- This interval is called the **level** $1-\alpha$ **confidence interval**.
- It is narrower the smaller the standard error.
- Thus, the smallness of the standard error is a measure of the estimator's precision.

Confidence interval

Interpretation - for a 95% confidence interval:

- If repeated sample is used to compute the confidence intervals, true value will be inside 95% of those confidence intervals.
- IT DOES NOT MEAN 95% probability of true value falling inside interval.

Note also, that confidence intervals provide the range of likely values of parameters for the population and not just point estimates.

p-Value

When we are rejecting the null, we do not differentiate between whether the test statistic is just outside of the rejection region, or way beyond the range. An alternative measure to look at is **p-value** which is the smallest significance level at which the null would be rejected given the observed value of test statistic

- So the decision rule of the t-test can also be stated using the **p-Value**
- Steps 1,2,3 same as above

Step 4': Rather than finding the critical value, calculate

$$p = P(t > |t_k|) * 2$$

Since the t distribution is symmetric around 0, $P(t < -|t_k|) = P(t > |t_k|)$,so

$$P(-|t_k| < t < |t_k|) = 1 - p \tag{10}$$

Step 5': Do not reject H_0 if $p > \alpha$. Otherwise, reject H_0 and accept H_1.

p-values

Few comments:

- Another interpretation: p -value is the probability of observing a t-statistic as extreme as we did if the null hypothesis is true
- p-value is a probability, so its value must be between 0 and 1
- p-value is most useful when rejection decision depends on choice of significance level